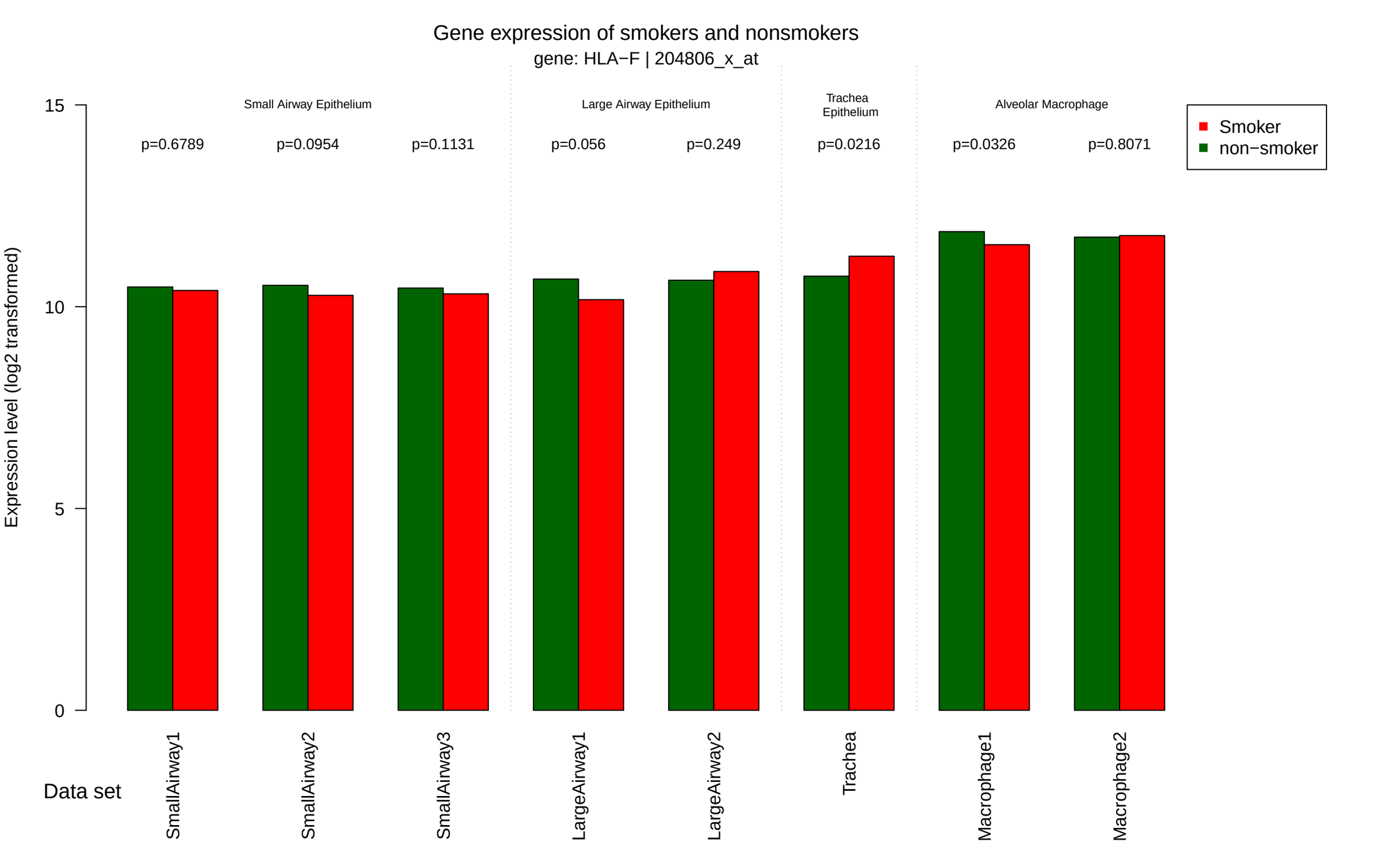
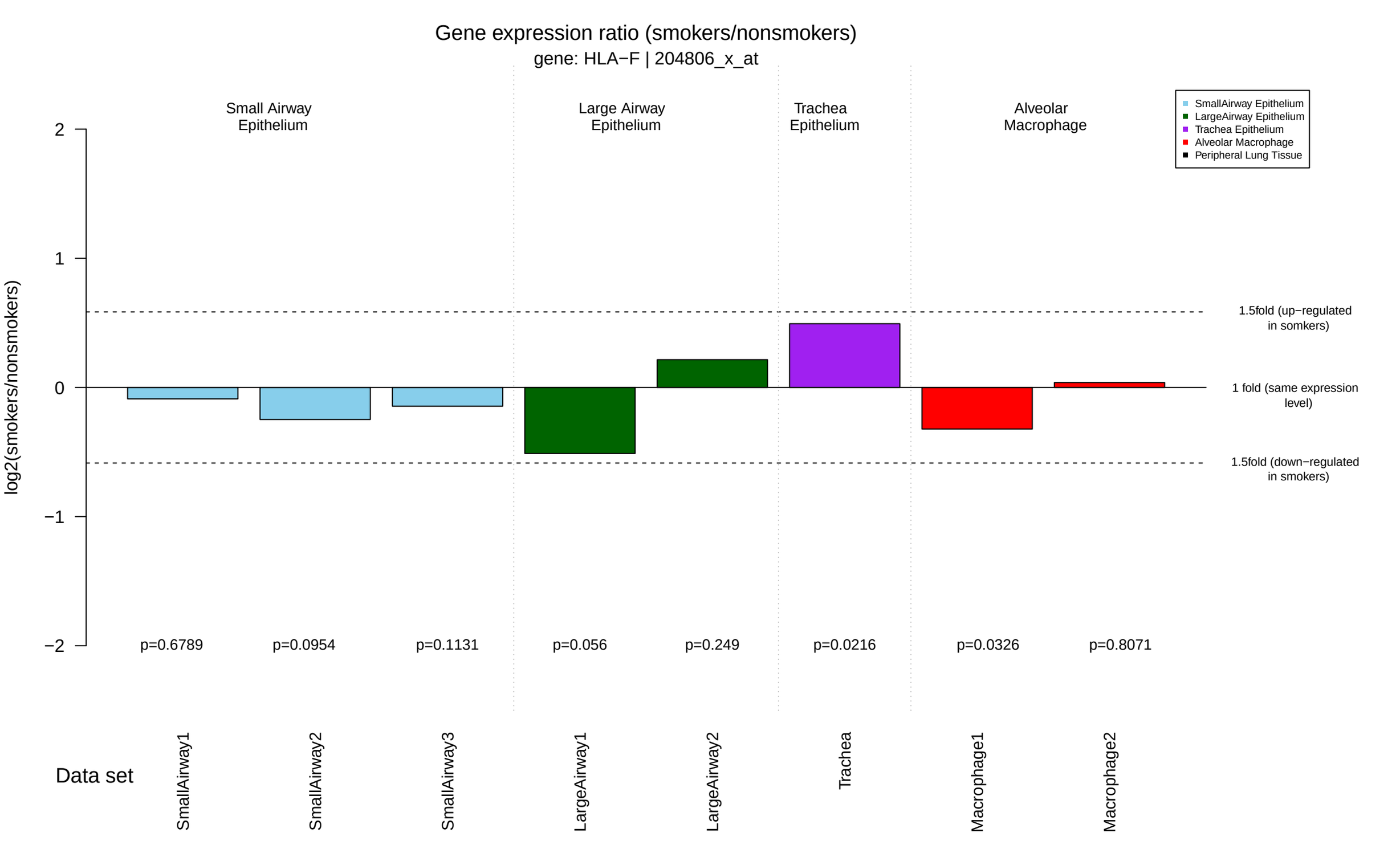
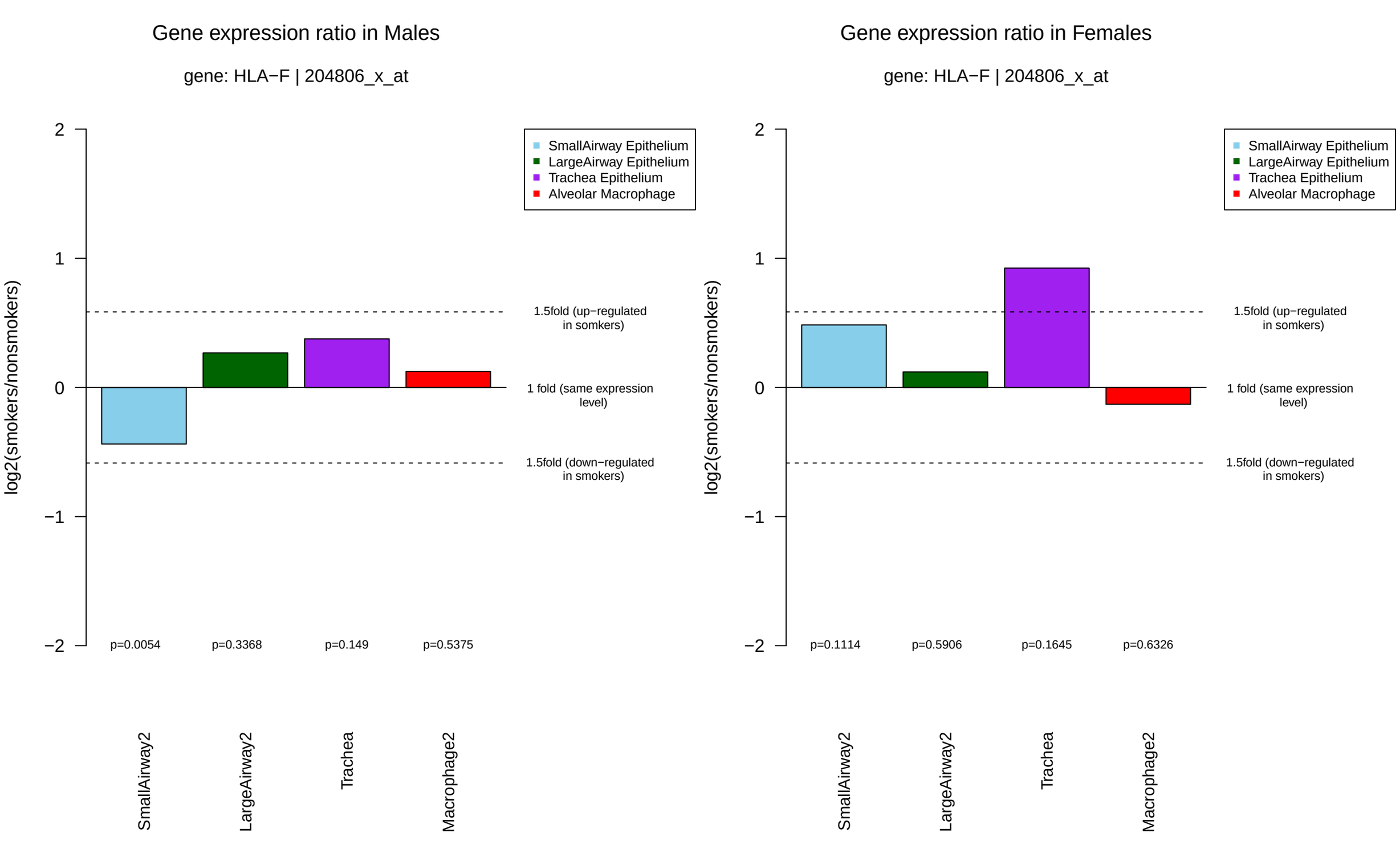
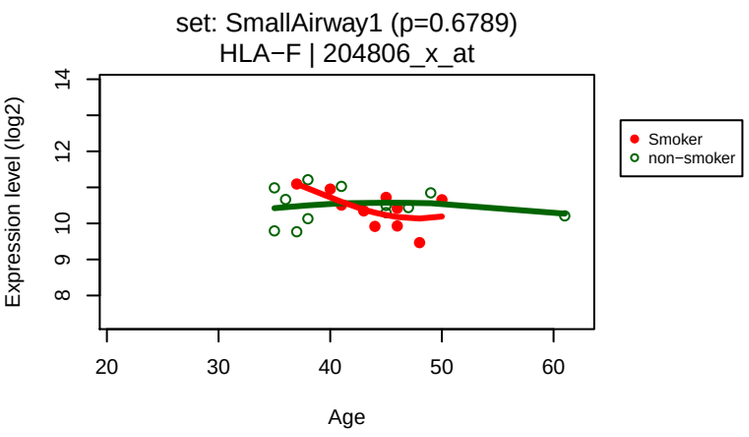
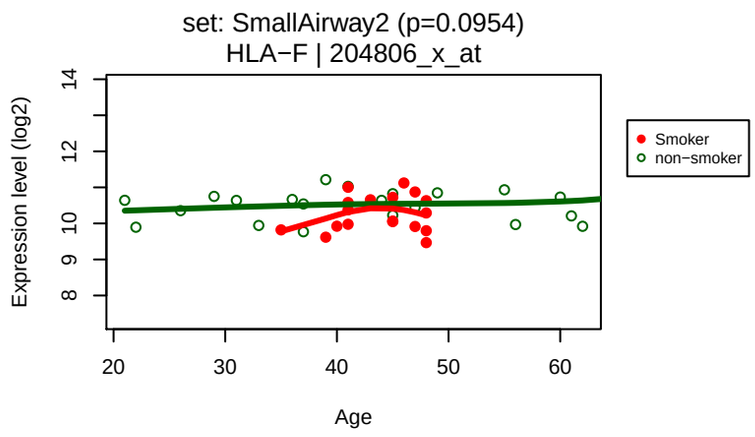
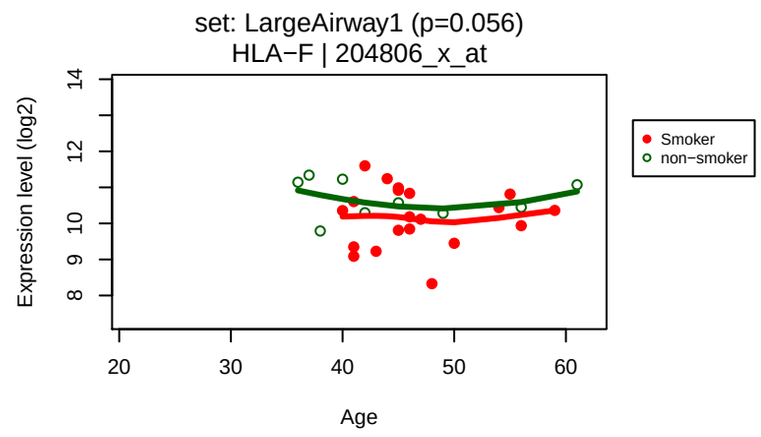

Gene expression of smokers and nonsmokers
gene: HLA−F | 204806_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6789
p=0.0954
p=0.1131
p=0.056
p=0.249
p=0.0216
p=0.0326
p=0.8071
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

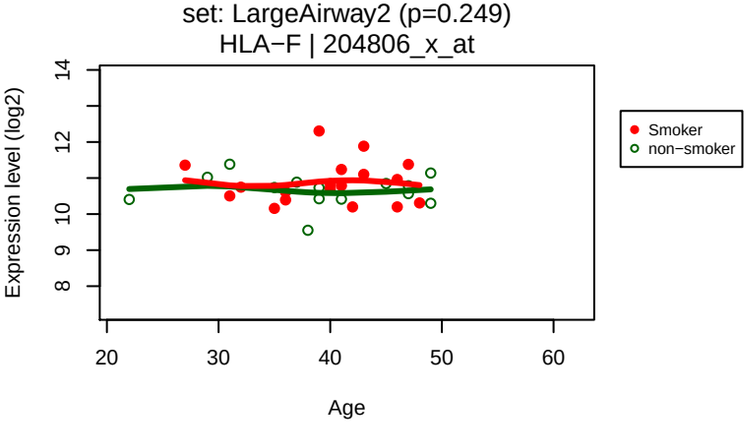

Gene expression ratio (smokers/nonsmokers)
gene: HLA−F | 204806_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6789
p=0.0954
p=0.1131
p=0.056
p=0.249
p=0.0216
p=0.0326
p=0.8071
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

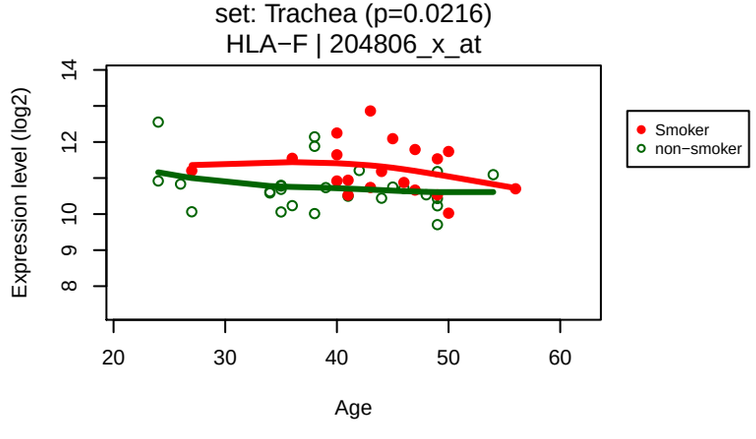

Gene expression ratio in Males
gene: HLA−F | 204806_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0054
p=0.3368
p=0.149
p=0.5375
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−F | 204806_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1114
p=0.5906
p=0.1645
p=0.6326
SmallAirway2
LargeAirway2
Trachea
Macrophage2

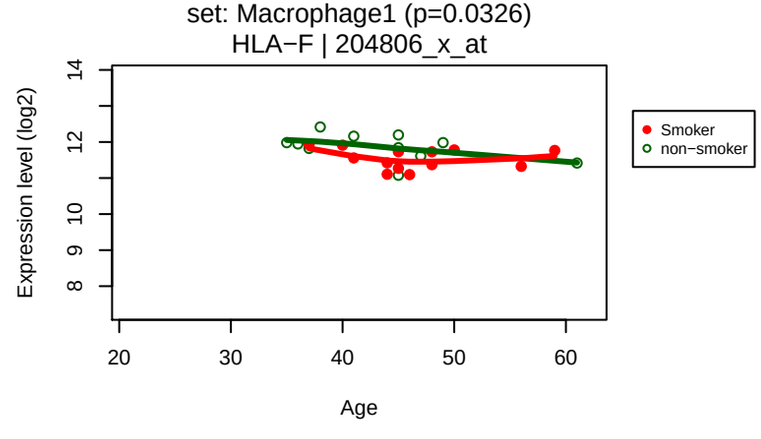

set: SmallAirway1 (p=0.6789)
HLA−F | 204806_x_at
Expression level (log2)
Age
Smoker
non−smoker

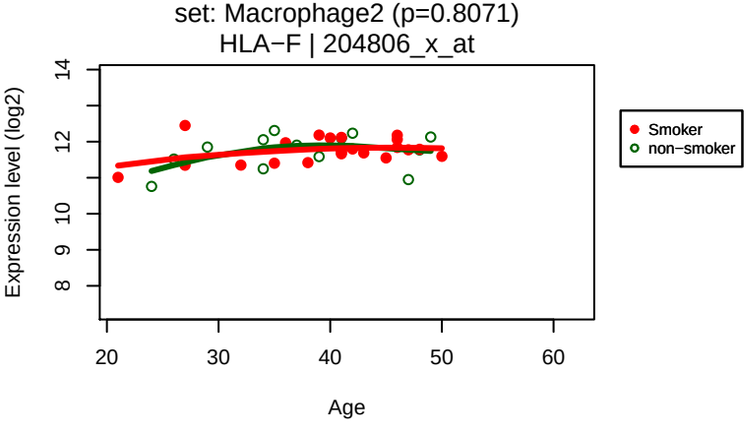

set: SmallAirway2 (p=0.0954)
HLA−F | 204806_x_at
Expression level (log2)
Age
Smoker
non−smoker

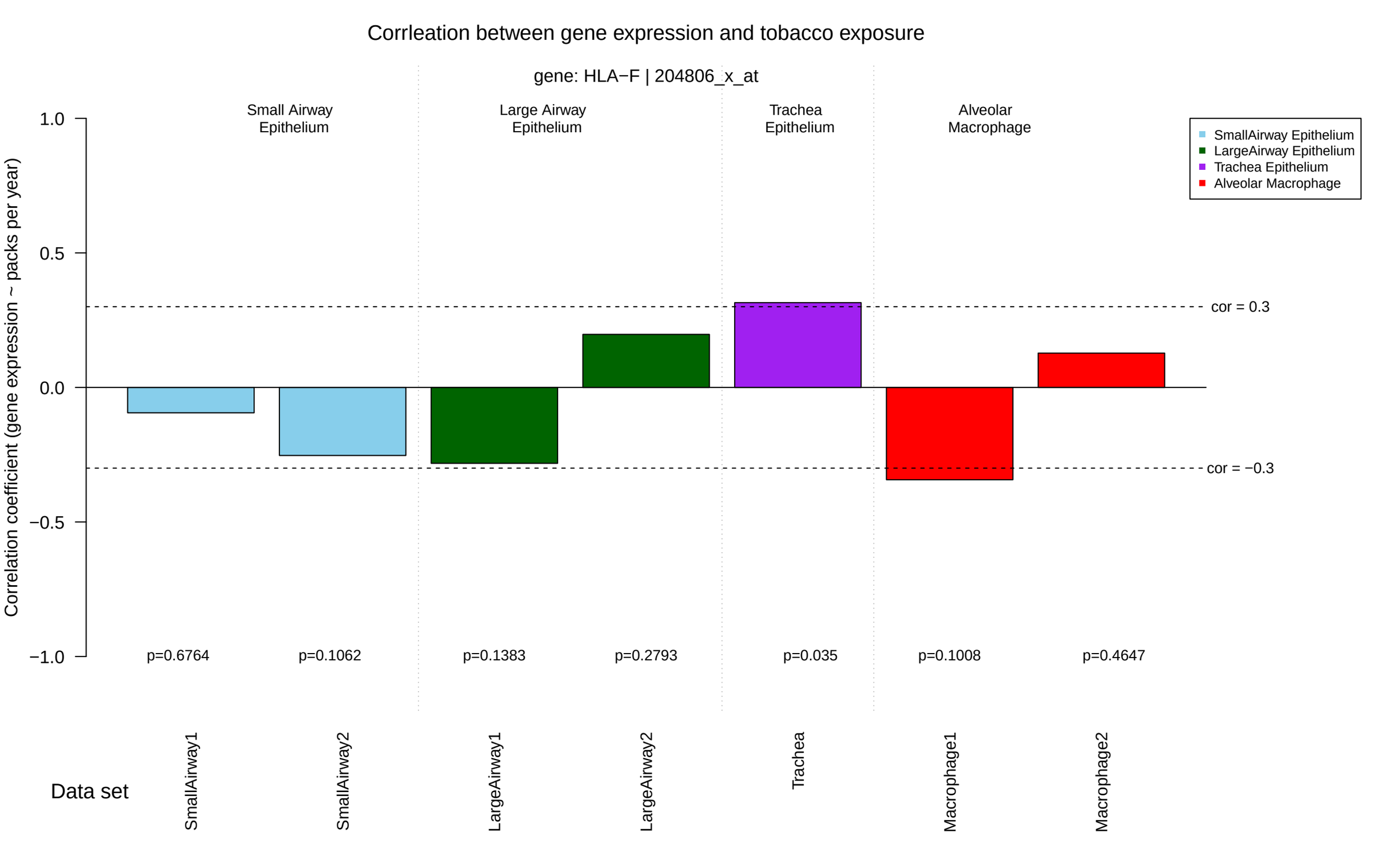

set: LargeAirway1 (p=0.056)
HLA−F | 204806_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.249)
HLA−F | 204806_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0216)
HLA−F | 204806_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0326)
HLA−F | 204806_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8071)
HLA−F | 204806_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−F | 204806_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6764
p=0.1062
p=0.1383
p=0.2793
p=0.035
p=0.1008
p=0.4647
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2